Style AI - AI-Powered Fashion Styling Advisor with Groq

Project Description

StyleAI is an intelligent fashion styling platform that leverages Groq's LLaMA 3.3 70B AI model to provide comprehensive personal styling recommendations. The platform addresses the challenge of personalized fashion guidance by delivering AI-powered styling insights, skin tone analysis, outfit recommendations, and curated shopping links.

Using Groq's advanced language model, StyleAI analyzes user photos, detects skin tone, and generates gender-specific fashion recommendations tailored to individual profiles. The system ensures fast response times through Groq's optimized API while maintaining high-quality styling advice through sophisticated image processing and AI analysis.

StyleAI transforms personal styling into an intelligent, user-friendly experience through its modern interface, comprehensive feature set, and AI-powered analysis that provides personalized fashion guidance while considering skin tone, gender preferences, and current fashion trends.

Scenarios

Scenario 1: Skin Tone Analysis & Initial Styling
A user uploads a clear facial photograph to StyleAI. The system detects their skin tone category (Fair, Medium, Olive, Deep), analyzes the facial region for accurate color detection, and provides initial personalized styling recommendations. The AI suggests color palettes, outfit combinations, and accessories that complement their specific skin tone.

Scenario 2: Gender-Specific Fashion Recommendations
A female user uploads her photo and indicates her gender preference. StyleAI generates tailored recommendations including:

- Suitable dress codes (Formal, Business, Casual, Party)
- Gender-specific outfit descriptions with tops, bottoms, and shoes
- Hairstyle suggestions with maintenance tips
- Accessory recommendations (earrings, necklaces, bracelets, watches)
- Color palette guidance (primary, secondary, accent colors)
- Detailed explanation of why recommendations work for her skin tone

Scenario 3: Curated Shopping Experience
After receiving styling recommendations, the user accesses curated product links from major Indian retailers (Amazon.in, Myntra, Zara). Products are specifically selected based on the user's skin tone and gender, including direct shopping links for shirts, pants, shoes, and accessories from verified e-commerce platforms.

Architecture Overview

StyleAI is built as a modular platform combining Flask backend with Groq's AI API for intelligent styling. The architecture prioritizes accuracy, speed, and user experience by leveraging advanced image processing and AI-driven recommendations.

Core Technologies

- **Flask**: Lightweight Python web framework for routing and request processing
- **Groq API**: Cloud-based AI inference using LLaMA 3.3 70B model
- **OpenCV**: Advanced image processing and face detection
- **PIL (Pillow)**: Image manipulation and color analysis
- **NumPy**: Numerical computation for color analysis
- **HTML/CSS/JavaScript**: Modern, responsive frontend with smooth animations

Pre-requisites

Software Requirements:

- Python 3.8+: Download Python (https://www.python.org/downloads/)
- Groq API Key: Obtain from https://console.groq.com
- Git: Download Git (https://git-scm.com/downloads)
- Code Editor: VS Code, PyCharm, or any preferred IDE

Knowledge Prerequisites:

- Python Basics: Functions, classes, exception handling
- Flask Framework: Flask Documentation & Routing
- HTML/CSS: Basic web development and styling
- JavaScript: DOM manipulation and fetch API
- Image Processing: Color models and face detection

Project Workflow

Phase 1: Environment Setup & Groq API Configuration

- Activity 1.1: Set up Python environment and install dependencies
- Activity 1.2: Obtain and configure Groq API key
- Activity 1.3: Test Groq API connectivity

Phase 2: Core Backend Development

- Activity 2.1: Set up Flask application structure
- Activity 2.2: Implement image upload and validation
- Activity 2.3: Implement skin tone detection algorithm
- Activity 2.4: Integrate Groq API for styling recommendations
- Activity 2.5: Develop product recommendation system

Phase 3: Frontend Development

- Activity 3.1: Design responsive HTML templates
- Activity 3.2: Implement image upload interface
- Activity 3.3: Create styling recommendations display

Phase 4: Deployment

- Activity 4.1: Local and production deployment

Phase 5: Testing & Optimization

- Activity 5.1: Functional testing and optimization

Technical Architecture & Project Structure Setup

App Entrypoint Code Screenshot (app.py):

app.py

```
from flask import Flask, render_template, request, jsonify
import numpy as np
import os
from styleai import create_app
from styleai.config import Config

app = create_app()
app.config['MAX_CONTENT_LENGTH'] = Config.MAX_CONTENT_LENGTH

if __name__ == '__main__':
  print('\n' + '=' * 70)
  print('🎨 STYLEAI FASHION ADVISOR - STARTING SERVER')
  print('=' * 70)
  app.run(host=Config.APP_HOST, port=Config.APP_PORT, debug=True)
```

MILESTONE 1: Environment Setup & Groq API Configuration

Establish the cloud AI infrastructure by configuring Groq API access and validating connectivity. This milestone ensures the AI backend is properly configured for generating styling recommendations.

Activity 1.1: Install Dependencies & Setup Environment

Create requirements.txt:

```
requirements.txt

Flask
Flask-CORS
Pillow
opencv-python
groq
python-dotenv
Werkzeug
```

Activity 1.2: Configure Groq API Key

Create .env file:

```
.env

FLASK_ENV =development
SECRET_KEY =dev-secret-key-change-in-production-12345
MAX_CONTENT_LENGTH_MB =10
UPLOAD_TMP_DIR =/tmp/styleai
GROQ_API_KEY =gsk_your_groq_api_key_here
PI_KEY =gsk_your_groq_api_key_here
GROQ_BASE_URL =https://api.groq.com/openai/v1
GROQ_MODEL =llama-3.3-70b-versatile
```

Obtain API Key Steps:

1. Visit https://console.groq.com
2. Sign up or login
3. Create API key in console
4. Copy key and paste in .env file

MILESTONE 2: Core Backend Development

Build the Flask-based backend infrastructure that handles image uploads, performs skin tone detection, and coordinates AI requests to Groq. This milestone creates the foundation for all StyleAI features.

Backend Libraries & Route Logic:

MILESTONE 3: Frontend Development - UI/UX Design

Create a modern, responsive user interface using HTML5, CSS3, and JavaScript that provides smooth image upload, real-time processing feedback, and styled recommendation display.

Activity 3.1: Main Interface (index.html)

Key Features:

- Image upload form with drag-and-drop support
- Gender selection (Male/Female)
- Real-time upload progress indicator
- Styled results display with recommendations and shopping links
- Responsive design for desktop and mobile
- Modern CSS with gradients and animations

Components:

1. Upload Section: File input with validation
2. Processing Section: Loading animation during AI analysis
3. Results Section: Skin tone display with RGB values, AI recommendations, Product cards with shopping links

index.html Template Code Screenshot:

templates/index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Styling AI - Your Personal Fashion Assistant</title>
  <link rel="stylesheet" href="static/css/styles.css">
</head>
<body>
  <nav class="navbar"><div class="logo">Styling AI</div></nav>
  <section id="home" class="hero">
    <h1>Your Personal AI Fashion Stylist</h1>
    <p>Upload your photo and get personalized styling recommendations</p>
  </section>
  <section id="upload" class="upload-section">
    <form id="style-form" enctype="multipart/form-data">
      <input type="radio" name="gender" value="Male"> Male
      <input type="radio" name="gender" value="Female" checked> Female
      <div id="dropzone">Drag & Drop Your Photo</div>
      <button type="submit">Generate Styling Guide</button>
    </form>
  </section>
</body>
</html>
```

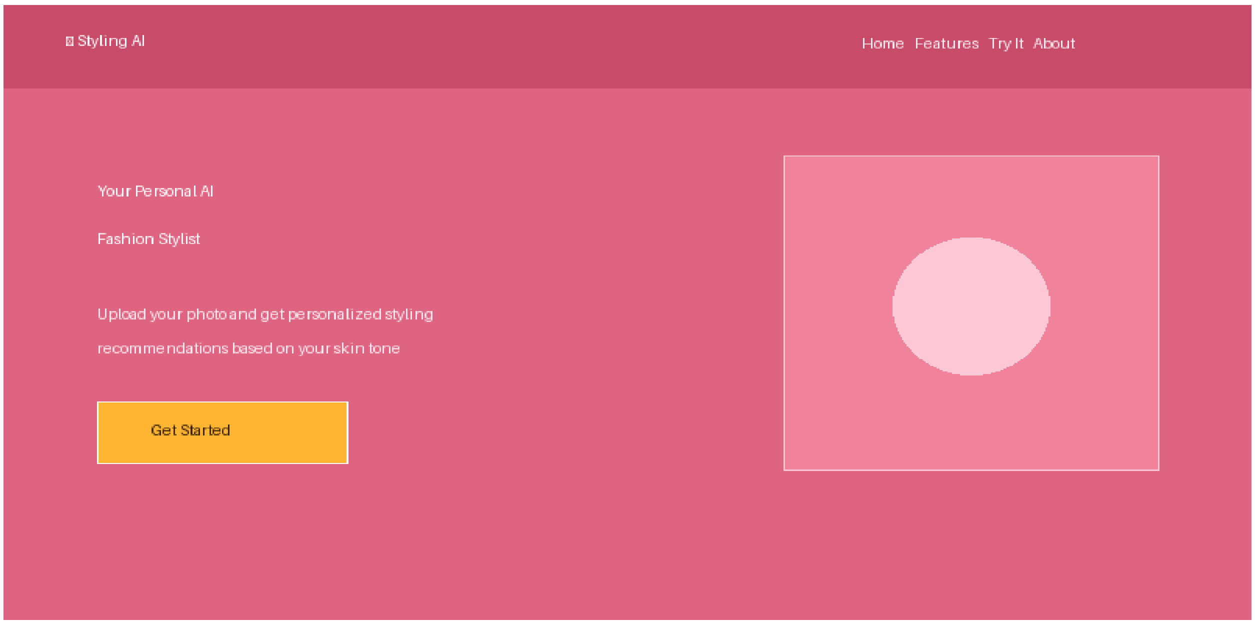
MILESTONE 4: Deployment

Activity 4.1: Local Deployment

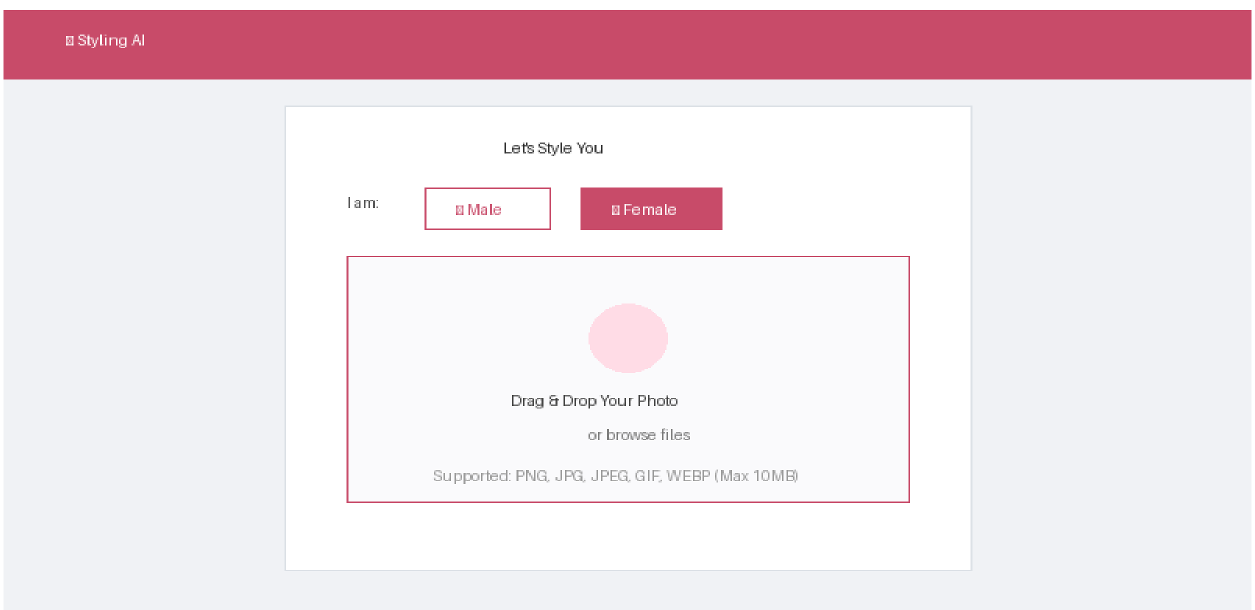
Run the application:

```
python app.py
```

Landing Page Hero Section UI Screenshot:



Upload Form & Gender Selector UI Screenshot:



MILESTONE 5: Testing & Optimization

Activity 5.1: Functional Testing

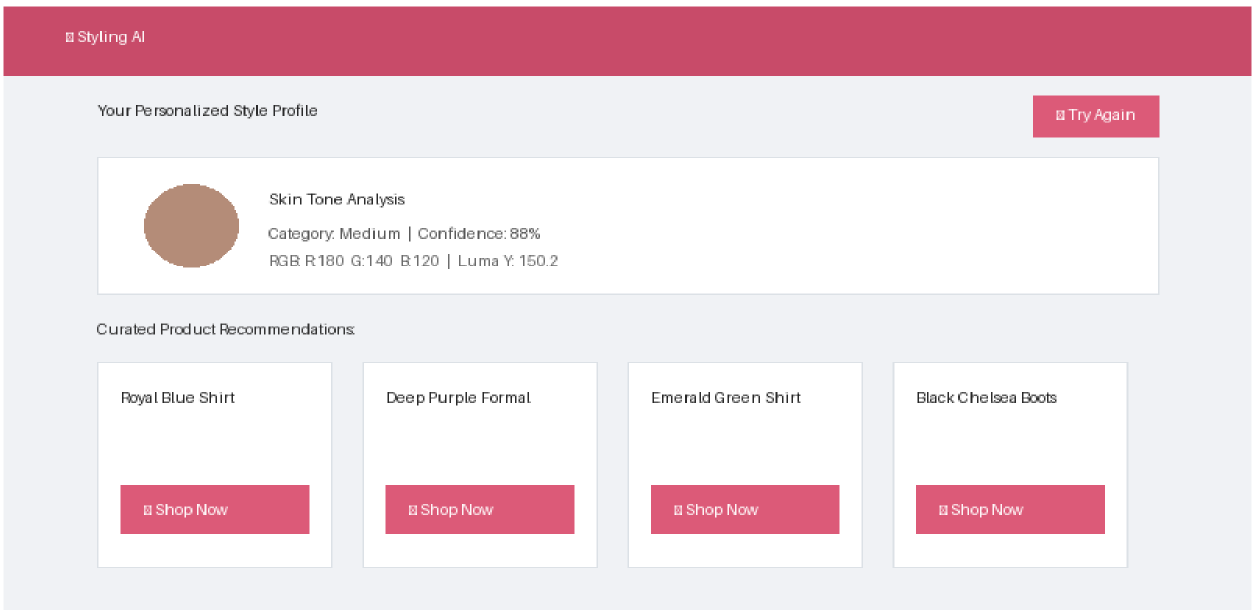
Test Case 1: Image Upload Validation

- Drag-and-drop & file browser input testing
- Validation of PNG, JPG, JPEG, GIF, WEBP format types

Test Case 2: Skin Tone Detection

1. Upload clear frontal facial photo
2. Select gender preference
3. Expected: Accurate skin tone detection with RGB values and styled recommendations

Personalized Style Profile & Curated Product Recommendations UI Screenshot:



Configuration Parameters

Parameter	Value
MAX_FILE_SIZE	10MB
ALLOWED_EXTENSIONS	png, jpg, jpeg, gif, webp
GROQ_MODEL	llama-3.3-70b-versatile
MAX_TOKENS	1200
TIMEOUT	90 seconds
TEMPERATURE	0.7

Potential Enhancements

- Expand skin tone categories beyond 4 (continuous color mapping)
- Add additional gender options (non-binary, prefer not to say)
- Database integration for dynamic product catalogs
- User accounts and styling history
- Multiple photo uploads for consistency
- Virtual try-on features
- Seasonal trend integration
- Style preference learning and personalization

Conclusion

StyleAI successfully delivers an intelligent personal styling assistant that combines advanced image processing with cutting-edge AI analysis. By leveraging Groq's LLaMA 3.3 70B model, the system provides fast, accurate, and personalized fashion recommendations based on skin tone analysis and user preferences. The platform demonstrates the viability of AI-powered fashion guidance, making personalized styling accessible to everyone while maintaining high performance and accuracy.